

■ Ride-Hailing SQL Project

Ola & Uber | PostgreSQL Case Study

Practice Questions — Data Cleaning & Analysis

Table	Key Columns	Rows
users	user_id, city, gender, age, signup_date	400
drivers	driver_id, rating, vehicle_type, total_trips	300
rides	ride_id, platform, ride_type, fare_amount, ride_status ...	400

SECTION A — Data Auditing

Q1. Audit the users table and count: NULL genders, NULL ages, negative ages, ages above 100, and NULL cities.

Hint: Use COUNT() FILTER (WHERE ...) for each condition in a single SELECT.*

Q2. Audit the rides table and count: negative fares, zero distances, negative driver earnings, surge multipliers below 1, and NULL ride statuses.

Hint: Similar FILTER pattern — run on the rides table.

Q3. Audit the drivers table: count invalid ratings (outside 1–5), negative total_trips, and NULL ratings.

Hint: rating > 5 OR rating < 0 catches out-of-range values.

SECTION B — Data Cleaning

Q4. Standardize the gender column in users: normalize 'M', 'MALE', 'male' → 'Male'; 'F', 'FEMALE', 'female' → 'Female'; 'OTHER' → 'Other'; everything else → 'Unknown'.

Hint: Use UPDATE ... SET gender = CASE WHEN UPPER(gender) IN (...) THEN ...

Q5. Replace all remaining NULL values in the gender column with 'Unknown'.

Hint: Simple UPDATE with WHERE gender IS NULL.

Q6. Fix inconsistent casing in the vehicle_type column of the drivers table (e.g. 'sedan', 'SEDAN' should become 'Sedan').

Hint: Use INITCAP() and only update rows where casing differs.

Q7. Set age to NULL for users where age is negative or greater than 100.

Hint: UPDATE ... SET age = NULL WHERE age < 0 OR age > 100.

Q8. Set driver rating to NULL for any driver where the rating is outside the valid range of 1.0 to 5.0.

Hint: WHERE rating < 1.0 OR rating > 5.0.

Q9. Fix negative total_trips in the drivers table by setting them to 0.

Hint: UPDATE ... SET total_trips = 0 WHERE total_trips < 0.

Q10. Delete rides where distance_km = 0, or fare_amount <= 0, or driver_earnings <= 0.

Hint: DELETE FROM rides WHERE ... OR ... OR ...

Note: These records represent data errors, not cancellations.

Q11. Identify duplicate rides defined as records sharing the same `user_id`, `driver_id`, `request_time`, and `pickup_location`.

Hint: GROUP BY those four columns HAVING COUNT() > 1.*

Q12. Remove duplicate rides, keeping only the row with the lowest `ride_id` in each duplicate group.

Hint: DELETE ... WHERE ride_id NOT IN (SELECT MIN(ride_id) FROM rides GROUP BY ...).

Q13. Find all cancelled rides that are missing a `cancel_reason`. Then update those records and set `cancel_reason` to 'Not_Specified'.

Hint: Two steps — first a SELECT to verify, then an UPDATE.

Q14. Add a computed (generated stored) column `fare_per_km` to the `rides` table, calculated as `fare_amount / distance_km` rounded to 2 decimal places. Return NULL when `distance_km = 0`.

Hint: Use ALTER TABLE ... ADD COLUMN ... GENERATED ALWAYS AS (...) STORED with a CASE WHEN.

Q15. Add a second computed column `driver_cut_pct` representing the percentage of the fare that goes to the driver, rounded to 1 decimal place.

*Hint: (driver_earnings / fare_amount) * 100, wrapped in a CASE WHEN fare_amount > 0.*

SECTION C — Basic Analysis

Q16. How many total rides exist for each `ride_status`? Sort by count descending.

Hint: GROUP BY ride_status, ORDER BY total DESC.

Q17. How many rides does each platform (Ola / Uber) have?

Hint: GROUP BY platform.

Q18. What is the average fare for Solo rides vs Shared rides? (Completed rides only.)

Hint: WHERE ride_status = 'Completed', GROUP BY ride_type.

Q19. How many users are registered in each city?

Hint: GROUP BY city on the users table.

Q20. How many drivers operate each vehicle type?

Hint: GROUP BY vehicle_type on the drivers table.

Q21. Which payment method is used most frequently for completed rides?

Hint: WHERE ride_status = 'Completed', GROUP BY payment_method, ORDER BY count DESC.

Q22. What is the total revenue generated each month? (Use `request_time` for month extraction.)

Hint: TO_CHAR(request_time, 'YYYY-MM') for month grouping.

Q23. What is the average driver rating for each vehicle type? Exclude NULL ratings.

Hint: WHERE rating IS NOT NULL, GROUP BY vehicle_type.

Q24. What is the average fare per km for Solo vs Shared rides? (Completed rides only.)

Hint: Use the fare_per_km computed column; GROUP BY ride_type.

SECTION D — Advanced Analysis

Q25. Compare Solo vs Shared rides across these metrics: total rides, average fare, average distance, average duration, average fare/km, total revenue, cancellation count, and cancellation rate (%). Include both Completed and Cancelled rides.

Hint: Use COUNT() FILTER (WHERE ride_status = 'Cancelled') for cancellation stats.*

Q26. Compare Ola vs Uber platform performance: total rides, average fare/km, total revenue, average fare, average driver earnings, cancellation count, and cancellation rate (%).

Hint: Same FILTER approach as Q25, GROUP BY platform.

Q27. Find the top 20 users by total spend on completed rides. Show: user_id, city, gender, age, total rides, total spend, average fare, maximum fare, Ola ride count, Uber ride count.

Hint: JOIN users with rides on user_id. Use FILTER for platform-specific counts.

Q28. Which city generates the highest total revenue from completed rides?

Hint: JOIN users → rides, GROUP BY city, SUM(fare_amount).

Q29. What percentage of rides on each platform are cancelled? Show the cancellation rate rounded to 1 decimal place.

*Hint: FILTER count / total count * 100.*

Q30. For each combination of platform and ride_type, find the average fare and total rides. Which combination is the most popular?

Hint: GROUP BY platform, ride_type.